Equation Sheet: Magnetism

Intro Physics II • PHYS 235 (calc) • Knight ch. 29–30

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals; **bonus** = not required — might appear as an extra-credit problem only.

Symbols and constants

$\vec{B}$	magnetic field (T = N/A·m)	I	current (A)	Φ_B	magnetic flux (Wb = T·m^2)
q	charge (C)	n	turns per unit length (1/m)	$\mathcal{E}$	induced emf (V)
$\vec{v}$	velocity (m/s)	N	number of turns	$\vec{\mu}$	magnetic dipole moment (A·m^2)
θ	angle between the two vectors	r, d	distance (m)	μ_0	$4\pi \times 10^{-7}$ T·m/A

$d\vec{l}$ the Biot–Savart differential: an infinitesimal piece of wire, carrying current I – the calculus limit of $\Delta\vec{L}$

1. The pattern: electric → magnetic (how to read the rest of this sheet)

insert a $\vec{v}\times$

Force on a charge
(same q; the field now has to be crossed with $\vec{v}$, not just multiplied)
$$\vec{F} = q\vec{E} \quad \longrightarrow \quad \vec{F} = q\vec{v} \times \vec{B}$$

Field of a point charge (Coulomb's law) → field of a *moving* point charge
(same $1/r^2$; direction is now set by the right-hand rule on $\vec{v} \times \hat{r}$, not just radially out)
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \quad \longrightarrow \quad \vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v} \times \hat{r}}{r^2}$$

the source swap: $q\vec{v} \to I\,\Delta\vec{L}$

A single moving charge generalizes to a continuous current
(this is exactly how the moving-point-charge law above becomes the Biot–Savart law, Sec. 2 – a current element is just a lot of moving charge packed into a short wire segment)
$$q\vec{v} \quad \longrightarrow \quad I\,\Delta\vec{L}$$

electric dipole → magnetic dipole

Dipole moment, torque, and orientation energy all carry straight across
($\vec{p} \to \vec{\mu}$, $\vec{E} \to \vec{B}$ – see Sec. 4 for the full set side by side)
$$\vec{p} \to \vec{\mu}\,, \qquad \vec{E} \to \vec{B}$$

2. Magnetic force on charges & currents (Knight 29.1–29.3, 29.7–29.8)

fundamental

Force on a moving charge
($\perp$ to both $\vec{v}$ and $\vec{B}$; use the right-hand rule; zero if $\vec{v} \parallel \vec{B}$; compare $\vec{F} = q\vec{E}$, Sec. 1)
$$\vec{F} = q\,\vec{v} \times \vec{B}\,, \qquad |\vec{F}| = |q|\,vB\sin\theta$$

Force on a straight current-carrying wire, length L
$$\vec{F} = I\,\vec{L} \times \vec{B}\,, \qquad |\vec{F}| = BIL\sin\theta$$

derived

Charge in a uniform field ($\vec{v} \perp \vec{B}$)
(circular motion; magnetic force does no work, so v is constant)
$$r = \frac{mv}{|q|B}$$

Period / cyclotron frequency
(independent of v and r)
$$T = \frac{2\pi m}{|q|B}\,, \qquad f = \frac{|q|B}{2\pi m}$$

Velocity selector (crossed $\vec{E}$, $\vec{B}$)
$$v = \frac{E}{B}$$

3. Sources of magnetic field (Knight 29.4–29.5)

fundamental

Field of a single moving point charge
(the magnetic twin of Coulomb's law, Sec. 1 – same $1/r^2$, direction by the right-hand rule on $\vec{v} \times \hat{r}$)
$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$

Chop a current into pieces, add up
(each short segment $\Delta\vec{L}$ of current I acts like a swarm of moving charges, $I\Delta\vec{L}$ standing in for $q\vec{v}$ – see Sec. 1)
$$\Delta\vec{B} = \frac{\mu_0}{4\pi} \frac{I\,\Delta\vec{L} \times \hat{r}}{r^2}, \qquad \vec{B} = \textstyle\sum_i \Delta\vec{B}_i$$

derived — given results, use these directly

Long straight wire (distance r)
(field circles the wire; $B \propto 1/r$)
$$B = \frac{\mu_0 I}{2\pi r}$$

Center of a circular loop (radius R, N turns)
$$B = \frac{\mu_0 N I}{2R}$$

Inside a long solenoid
(uniform; independent of radius; ≈ 0 outside)
$$B = \mu_0 n I$$

4. Ampère's law — the idea (Knight 29.6)

fundamental

Only the enclosed current matters
(any closed loop around a current; current entirely outside the loop contributes nothing; sign by the right-hand rule)
$$I_{\text{enc}} = \textstyle\sum_i I_i \quad \text{(signed)}$$

derived — apply it with the Sec. 2 results

Straight wire (distance r)
$$B = \frac{\mu_0 I}{2\pi r} \quad \text{(Sec. 2 — Ampère's law is how it's derived)}$$

Inside a long solenoid
$$B = \mu_0 n I \quad \text{(Sec. 2, same way)}$$

5. Magnetic dipoles & torque (Knight 29.5, 29.9)

fundamental

Dipole moment of a current loop
(compare the electric dipole moment $\vec{p} = q\vec{d}$)
$$\vec{\mu} = NI\vec{A} \qquad (|\vec{\mu}| = NIA)$$

Torque on a dipole in a field
(twists $\vec{\mu}$ toward $\vec{B}$; max at $\theta = 90°$, zero when aligned; compare $\vec{\tau} = \vec{p} \times \vec{E}$)
$$\vec{\tau} = \vec{\mu} \times \vec{B}, \qquad |\vec{\tau}| = \mu B \sin\theta$$

derived

Orientation energy
(minimum when $\vec{\mu} \parallel \vec{B}$; compare $U = -\vec{p} \cdot \vec{E}$)
$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$$

Net force on a dipole in a *uniform* field
$$\vec{F}_{\text{net}} = 0 \quad \text{(torque only)}$$

6. Magnetic flux & electromagnetic induction (Knight 30.1–30.7)

fundamental

Magnetic flux through a loop (uniform field: just area times the component of $\vec{B}$ along the normal)	$\Phi_B = BA\cos\theta$
Faraday's law — induced emf (only a *changing* flux induces an emf)	$\mathcal{E} = -N\dfrac{\Delta\Phi_B}{\Delta t}$
Lenz's law	induced current opposes the change in flux

derived

Motional emf (bar of length L, speed v, $\vec{v}\perp\vec{B}$)	$\mathcal{E} = BLv$
Induced current and dissipated power	$I = \dfrac{\mathcal{E}}{R}, \qquad P = \dfrac{\mathcal{E}^2}{R} = F_{\text{pull}}\,v$
Induced electric field (inside a region of changing B, radius r from center) (circulates even with no wire present; direction by Lenz's law, as if a loop were there)	$E(r) = \dfrac{r}{2}\left\lvert\dfrac{dB}{dt}\right\rvert$

7. Doing the calculus: Biot–Savart & Ampère's law (Knight 29.4–29.6)

fundamental

Biot–Savart law — field of a current element (the calculus limit of Sec. 2's chop-and-add rule: $\Delta\vec{L}\to d\vec{l}$, sum $\to$ integral)	$d\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{I\,d\vec{l}\times\hat{r}}{r^2}, \qquad \vec{B} = \int d\vec{B}$
Ampère's law (any closed loop; I_{enc} is the *signed* current through the loop – this is the formal version of Sec. 3's "only enclosed current matters")	$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{\text{enc}}$

derived

Evaluating an Amperian-loop integral (pick a loop where B is constant and $\parallel$ to $d\vec{l}$ everywhere)	$\oint \vec{B}\cdot d\vec{l} = B\oint dl = B\,L_{\text{loop}} \;\Rightarrow\; B = \dfrac{\mu_0 I_{\text{enc}}}{L_{\text{loop}}}$
Current entirely outside the loop	$\oint \vec{B}\cdot d\vec{l} = 0$

8. Bonus: torque and orientation energy might appear as an extra-credit problem only

torque ↔ energy

Torque is the (negative) slope of U with respect to angle (same relationship as $F_x = -dU/dx$ on the electrostatics sheet, just swap position for angle)	$\tau_\theta = -\dfrac{dU}{d\theta}$

Check it: $U = -\mu B\cos\theta \Rightarrow -dU/d\theta = -\mu B\sin\theta$, which matches $|\vec{\tau}| = \mu B\sin\theta$ (Sec. 4) up to the sign convention for which way θ increases.

Companion simulations at `hoppese.github.io/Intro-physics-sims/`: `moving-charge-field`, `magnetic-field-of-currents`, `discrete-current-builder-3d`, `ampere-loop`, `dipole-field-and-torque`, `lorentz-force`, `current-loop-motor`, `faraday-induction`, `faraday-flux-explorer`, `rail-flux-bar`.